Xinyan Wang

Email : **xwang2587@wisc.edu**

Personal Website: **https://xinyan-wang-stat.github.io/**

Linkedin: **xinyan-wang-34b462206**

Research Interests

LLM post-training (RL, on-policy distillation), LLM agents and agent red-teaming, efficient inference.

Education

- **University of Wisconsin–Madison** — Madison, WI, USA
  Ph.D. in Statistics; Advisor: Prof. Jun Shao — *Sep 2023 – May 2028 (Expected)*
  M.S. in Computer Science — *Sep 2023 – May 2026*
  M.S. in Statistics — *Sep 2022 – May 2023*
  Visiting Student in Statistics and Computer Science — *Sep 2021 – May 2022*
- **East China Normal University** — Shanghai, China
  B.S. in Statistics — *Sep 2018 – June 2022*

Experience

- **SaFo Lab** — Johns Hopkins University
  Research Intern; Mentor: Prof. Chaowei Xiao — *Aug 2025 – Present*
  - Research on safe and efficient large reasoning models and LLM agents.

Research Experience

- **ROM: Real-time Overthinking Mitigation in Large Reasoning Models** — JHU
  First author; advised by Prof. Chaowei Xiao (arXiv 2026)
  - Framed overthinking in large reasoning models (LRMs) as a latent productive-to-redundant transition, linearly decodable from late-layer hidden states around first-correct-solution (FCS) boundaries.
  - Built ROM, a streaming framework that monitors a frozen LRM with a lightweight detector ($\sim$0.1% of parameters) and intervenes at reasoning boundaries in real time; Counterfactual Self-Correction (CSC) preserves pre-FCS self-correction.
  - Across five backbones, five benchmarks, and ten baselines, attained the highest accuracy in 19 of 25 settings, cut response length 28–77% (mean 45%), and cut latency 46.5% via zero-shot transfer.
- **ReasoningBomb: A Denial-of-Service Attack on Large Reasoning Models** — JHU
  With Prof. Chaowei Xiao et al. (ACM CCS 2026)
  - Formalized prompt-induced inference-time denial-of-service (PI-DoS) for LRMs and built a reinforcement-learning attacker that crafts short, natural prompts inducing pathologically long reasoning.
  - Across 7 open-source and 3 commercial models, induced 286.7× input-to-output amplification and bypassed input-, output-, and dual-stage detectors at 99.8%, 98.7%, and 98.4%.
- **PW-OPSD: Position-Weighted On-Policy Self-Distillation for Reasoning** — JHU
  With Prof. Chaowei Xiao et al. (arXiv 2026)
  - Built a branch-viability diagnostic that forces privileged-teacher token alternatives after the student's on-policy prefix and tests whether the correct answer still emerges.
  - Showed on Qwen3-4B that within-sequence position predicts teacher-token reliability far better than local uncertainty (AUROC 0.83 vs. 0.57); the resulting PW-OPSD up-weights later tokens, improving AIME 2024/2025 Avg@12 by +1.0/+1.1 at no extra teacher compute.
- **MLE with Datasets from Populations Having Shared Parameters** — UW–Madison
  Advisor: Prof. Jun Shao (Statistical Theory and Related Fields, 2023)
  - Studied maximum likelihood estimation with multiple datasets from heterogeneous populations sharing parameters; established asymptotic normality, estimated variability via bootstrap, and reduced the estimator's standard deviation by nearly 57% versus single-dataset MLEs.

Publications

Xiaogeng Liu, **Xinyan Wang**, Yechao Zhang, Sanjay Kariyappa, Chong Xiang, Muhao Chen, G. Edward Suh, Chaowei Xiao. *ReasoningBomb: A Stealthy Denial-of-Service Attack by Inducing Pathologically Long Reasoning in Large Reasoning Models.* ACM Conference on Computer and Communications Security (CCS), 2026.

Xinyan Wang, Xiaogeng Liu, Ming Pei, Chaowei Xiao. *ROM: Real-time Overthinking Mitigation via Streaming Detection and Intervention.* arXiv:2603.22016, 2026.

Xiaogeng Liu, **Xinyan Wang**, Yingzi Ma, Yechao Zhang, Chaowei Xiao. *When Are Teacher Tokens Reliable? Position-Weighted On-Policy Self-Distillation for Reasoning.* arXiv:2605.21606, 2026.

Jun Shao, **Xinyan Wang**. *MLE with Datasets from Populations Having Shared Parameters.* Statistical Theory and Related Fields, 7(3):213–222, 2023.

Teaching

STAT 240 (Data Science Modeling I), STAT 309 (Introduction to Probability and Mathematical Statistics I), STAT 340 (Data Science Modeling II), STAT 610 (Statistical Inference), STAT 611 (Statistical Models for Data Science), STAT 613 (Statistical Methods for Data Science).

Professional Service

Reviewer: ACL 2026, ECCV 2026, EMNLP 2026, NeurIPS 2026

Skills

Languages: Python, R, SQL
Frameworks: PyTorch, Scikit-learn, Flask